Lecture Notes

on Vector Dynamics, and Matrix Algebra

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1 Vector Dynamics

1.1 Sequence of Vectors

Just as you can have a sequence of numbers, you can also have a **sequence of vectors**.

Example: Consider the following list of concentrations of sodium (Na^+) and chlorine (Cl^-) ions recorded at successive time steps.

Time	Na^+ (mol/L)	Cl^- (mol/L)	Vector Representation
$t = 1$	1.63	2.63	$\begin{pmatrix} 1.63 \\ 2.63 \end{pmatrix} = \vec{v}(1)$
$t = 2$	1.83	2.86	$\begin{pmatrix} 1.83 \\ 2.86 \end{pmatrix} = \vec{v}(2)$
$t = 3$	1.95	2.95	$\begin{pmatrix} 1.95 \\ 2.95 \end{pmatrix} = \vec{v}(3)$
$t = 4$	1.99	2.99	$\begin{pmatrix} 1.99 \\ 2.99 \end{pmatrix} = \vec{v}(4)$

Here, $f(t)$ and $g(t)$ denote the sodium and chlorine concentrations, respectively, and we can express each measurement as a 2-dimensional vector:

$$\vec{v}(t) = \begin{pmatrix} f(t) \\ g(t) \end{pmatrix}.$$

1.2 Interpreting Rows and Columns

Each **row** in the table corresponds to a different time step:

$$\text{Row 1: } t = 1, \quad \text{Row 2: } t = 2, \quad \text{Row 3: } t = 3, \quad \text{Row 4: } t = 4.$$

Each **column** represents a distinct quantity:

$$\text{Column 1: Time}, \quad \text{Column 2: } f(t)\ (\text{Na}^+), \quad \text{Column 3: } g(t)\ (\text{Cl}^-), \quad \text{Column 4: } \vec{v}(t).$$

Thus, the collection of all $\vec{v}(t)$ forms a **vector sequence**:

$$\vec{v}(1),\ \vec{v}(2),\ \vec{v}(3),\ \vec{v}(4),\ \ldots$$

This provides a compact way to represent the joint evolution of multiple related quantities, such as concentrations, positions, or populations, over time.

2 Matrix

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}_{3\times1}$$

2.1 Definition

A **matrix** is a two-dimensional array of numbers. An $m \times n$ matrix (read as "m by n matrix"), say A, has m rows and n columns.

$$\begin{pmatrix} 2 & 3 & 1 \\ 0 & 2 & 5 \end{pmatrix}_{2\times3}$$

If a_{ij} denotes the entry in the i^{th} row and j^{th} column, then

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m\times n}.$$

$$\begin{pmatrix} 2 & 3 \\ 5 & 1 \end{pmatrix}_{2\times2}$$

Here, the diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$ form the **main diagonal** of the matrix.

Row i : the i^{th} horizontal line of entries, Column j : the j^{th} vertical line of entries.

2.2 Square Matrix

If for a matrix the number of rows equals the number of columns $(m = n)$, then the matrix is called a **square matrix**.

☛ **Example:** For the matrix $A_{m\times n}$ above, A will be a square matrix if $m = n$.

2.3 Diagonal Matrix

A **diagonal matrix** is a square matrix in which all the entries other than those on the main diagonal are zero. The entries on the main diagonal may be zero or nonzero.
Formally,

$$D_{m\times n} = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} = \operatorname{diag}(d_1, d_2, \dots, d_n).$$

☛ **Example:**

$\begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}$ is a square matrix but not a diagonal matrix.

$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is a diagonal matrix.

2.4 Identity Matrix

If the main diagonal entries of a diagonal matrix are all 1, then the matrix is called an **identity matrix**, denoted by $I_{m\times n}$.

$$I_{2\times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I_{4\times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

2.5 Zero Matrix

If all entries of a matrix are zero, then the matrix is called a **zero matrix**, denoted by

$$O_{m\times n} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}.$$

3 Matrix Operations

3.1 Matrix Addition

Let A and B be two matrices of the same size $m \times n$. Then one can perform the **matrix addition** $A + B$, defined as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & b_{m3} & \dots & b_{mn} \end{bmatrix}.$$

The sum $A + B$ is obtained by adding the corresponding entries:

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}_{m\times n}.$$

Example:

$$A = \begin{bmatrix} 2 & 4 & 8 \\ 6 & 3 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 3 \\ 4 & 1 \\ 0 & 1 \end{bmatrix}.$$

$$A \in \mathbb{R}^{2\times 3}, \quad B \in \mathbb{R}^{2\times 3}, \quad C \in \mathbb{R}^{3\times 2}.$$

Can you perform $A + C$? No. Matrix addition is defined only when both matrices are of the same size.

$$A + B = \begin{bmatrix} 2+2 & 4+0 & 8+1 \\ 6+3 & 3+4 & 7+2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 9 \\ 9 & 7 & 9 \end{bmatrix}.$$

3.2 Scalar Multiplication

If $c \in \mathbb{R}$, then the scalar multiplication of a matrix A by c is defined as

$$cA = \begin{bmatrix} ca_{11} & ca_{12} & \dots & ca_{1n} \\ ca_{21} & ca_{22} & \dots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \dots & ca_{mn} \end{bmatrix}.$$

That is, each entry of A is multiplied by the scalar c.

$A_{2\times 3}$

$B_{4\times 2}$

3.3 Matrix Multiplication

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then the **matrix product** AB is defined and will be of size $m \times p$. The product AB is obtained by multiplying each row of A with each column of B.

$C_{3\times 4}$

AB no

AC yes

$\Downarrow$

size

2×4

Important:

- The multiplication AB is defined only when the number of columns of A equals the number of rows of B.
- The product BA may not be defined if the sizes are incompatible.
- Even if both AB and BA exist (for example, if both are square matrices of size $m \times m$), in general,

  $$AB \neq BA.$$

  Thus, **matrix multiplication is not commutative.**

4 Illustration of Matrix Multiplication

Consider two matrices:

$$(u_{11} \quad u_{12}) \cdot \begin{pmatrix} w_{11} \\ w_{21} \end{pmatrix}$$

$$(u_{11} \quad u_{12}) \cdot \begin{pmatrix} w_{12} \\ w_{22} \end{pmatrix}$$

$$U = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}_{2\times 2}, \qquad W = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}_{2\times 3}.$$

Can you perform UW? Yes, size = 2×3 $\quad WU = ?$ No.

What is the size of UW?

$$R = UW = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \end{bmatrix}_{2\times 3}.$$

$r_{11} = u_{11} w_{11} + u_{12} w_{21}$

$r_{12} = u_{11} w_{12} + u_{12} w_{22}$

$r_{23} = u_{21} w_{13} + u_{22} w_{23}$.

4.1 Computation of Each Entry

Each entry r_{ij} in the product matrix R is obtained by taking the **dot product** of the i^{th} row of U with the j^{th} column of W:

$$r_{ij} = \sum_{k=1}^{2} u_{ik} w_{kj}.$$

Explicitly:

$$\begin{aligned}
r_{11} &= u_{11}w_{11} + u_{12}w_{21},\\
r_{12} &= u_{11}w_{12} + u_{12}w_{22},\\
r_{13} &= u_{11}w_{13} + u_{12}w_{23},\\[4pt]
r_{21} &= u_{21}w_{11} + u_{22}w_{21},\\
r_{22} &= u_{21}w_{12} + u_{22}w_{22},\\
r_{23} &= u_{21}w_{13} + u_{22}w_{23}.
\end{aligned}$$

4.2 Visual Interpretation

Step 1. Multiply each element of the row of U with the corresponding element of the column of W.
Step 2. Add these products to obtain the corresponding entry in R.

$$\begin{bmatrix} u_{11} & u_{12} \end{bmatrix} \cdot \begin{bmatrix} w_{11} \\ w_{21} \end{bmatrix} = r_{11}, \qquad \begin{bmatrix} u_{11} & u_{12} \end{bmatrix} \cdot \begin{bmatrix} w_{12} \\ w_{22} \end{bmatrix} = r_{12}, \qquad \text{and so on.}$$

4.3 Compact Representation

$$R = \begin{bmatrix} u_{11}w_{11} + u_{12}w_{21} & u_{11}w_{12} + u_{12}w_{22} & u_{11}w_{13} + u_{12}w_{23} \\ u_{21}w_{11} + u_{22}w_{21} & u_{21}w_{12} + u_{22}w_{22} & u_{21}w_{13} + u_{22}w_{23} \end{bmatrix}.$$

Hence, the product UW combines rows of U and columns of W to form each entry in R, one **dot product** at a time.

Example:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}_{2\times 3}, \qquad B = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}_{3\times 2}.$$

Can you perform AB? Yes, Size = 2×2

Can you perform BA? Yes, Size = 3×3.

$$AB = \begin{bmatrix} (1\times 3 + 0\times 2 + 2\times 1) & (1\times 1 + 0\times 1 + 2\times 0) \\ (-1\times 3 + 3\times 2 + 1\times 1) & (-1\times 1 + 3\times 1 + 1\times 0) \end{bmatrix}$$

$$\Rightarrow AB = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} (3\times1 + 1\times-1) & (3\times0 + 1\times3) & (3\times2 + 1\times1) \\ (2\times1 + 1\times-1) & (2\times0 + 1\times3) & (2\times2 + 1\times1) \\ (1\times1 + 0\times-1) & (1\times0 + 0\times3) & (1\times2 + 0\times1) \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 & 7 \\ 1 & 3 & 5 \\ 1 & 0 & 2 \end{bmatrix}$$

For this example of matrices pairs A, B, $AB \neq BA$ i.e, A and B are not commutative.

5 Properties of Matrix Addition

Let A, B, C be $m \times n$ matrices and $a, b \in \mathbb{R}$. Then the following properties hold:

1. $A + B = B + A$ (Commutative Law)
2. $A + 0 = 0 + A = A$ (Additive Identity)
3. $A + (-A) = (-A) + A = 0$ (Additive Inverse)
4. $(A + B) + C = A + (B + C)$ (Associative Law)
5. $(a + b)A = aA + bA$ (Distributive over Scalars)
6. $a(A + B) = aA + aB$ (Distributive over Matrix Addition)

6 Properties of Matrix Multiplication

Let A, B, C be matrices such that the following matrix multiplications are defined, and let $a, b \in \mathbb{R}$. Then:

1. $A(BC) = (AB)C$ (Associative Law)
2. Matrix multiplication is **not necessarily commutative**, i.e., in general,
$$AB \neq BA.$$
3. $A(B + C) = AB + AC$ (Distributive Law)
4. $(B + C)A = BA + CA$ (Distributive Law)
5. If A is an $m \times n$ matrix and I is the $n \times n$ identity matrix, then
$$AI = A.$$
Furthermore, if $m = n$, then
$$AI = IA = A.$$
6. $(aA)(bB) = (ab)(AB)$ (Compatibility of Scalar and Matrix Multiplication)

7 Matrix Representation of a Linear System of Equations

Consider the system:

$$\begin{cases} a_1x + a_2y + a_3z = d_1, \\ b_1x + b_2y + b_3z = d_2, \\ c_1x + c_2y + c_3z = d_3. \end{cases}$$

This system can be represented in **matrix form** as:

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}.$$

That is,

$$A\vec{x} = \vec{d},$$

where

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \vec{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}.$$